

The WITS Geeks

Project Plan: SA Learnerships and Skills Development

Overview

- This project involves the development of a Youth Leadership Skills and Development Portal, which is a web-based platform aimed at addressing youth unemployment in South Africa. The system connects job seekers with learnerships, and job opportunities posted by Providers. Administrators oversee the platform by approving opportunities and managing user roles.
 - The development follows an Iterative Scrum Methodology and is divided into four sprints. Each sprint delivers incremental, testable functionality and concludes in a fully featured portal with analytics, notifications, and automated matching.
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Technology Stack

Frontend

- HTML
- CSS
- JavaScript

Backend

- Node.JS
- Express.Js

Database

- MongoDB inc
- MongoDB Compass

Authentication

- Google as third-party provider + JWT (JSON Web Tokens)

Version Control

- GitHub (feature branches, pull requests)

CI/CD

- GitHub Actions, Azure deployment

Testing

- Jest (code coverage tool)

Design

- Figma

API Testing

- Postman

Deployment and Infrastructure

- Hosting Platform: Microsoft Azure
- Deployment Strategy: Automatic deployment of changes that pass tests
- Environment Variables: Managed separately for development and testing
- Repository: GitHub organization created with all team members granted access

Major Milestones and Deadlines

Spring 1

Duration: 2 weeks (28 March – 13 March 2026)

Milestone	Deadline
Sprint Planning and Client Meeting	28 March 2026
User story refinement and Task Allocation	01 April 2026
GitHub and Project Setup Structure	01 April 2026
Code Progress Review	08 April 2026
Middleware and Unfinished Features Assignment	09 April 2026
All Sprint 1 Tasks Completed	13 April 2026

Deliverables

- Google authentication (registration and login)
- Default application role assignment
- Role based access control (Applicant, Provider and Administrator)
- Provider ability to post opportunity listing
- Admin ability to approve/reject opportunities
- Basic dashboards per role

Spring 2

Duration: 1 weeks (14 April – 19 April 2026)

Milestone	Deadline
Sprint 2 Planning and Task Allocation	14 April 2026
Progress Update Meeting	16 April 2026
Progress Update Meeting and Issue Tracker	17 April 2026
Code Progress Review and Sprint Progress Check	19 April 2026
All Sprint 2 Tasks Completed	19 April 2026

Deliverables

- Providers can view opportunities they created
- Resubmit rejected opportunities
- Applicants can create and edit profiles
- Submit applications linked to profiles
- Track application statuses
- Unit tests with minimum 65% code coverage
- UML diagrams

Spring 3

Duration: 1 weeks (05 May – 10 April 2026)

Milestone	Deadline
Sprint 3 Planning ,Tension Resolution and Task Allocation	05 May 2026
Progress Update Meeting	07 May 2026
Progress Update Meeting	08 May 2026
Code Progress Review and Sprint Progress Check	09 May 2026
All Sprint 3 Tasks Completed	10 May 2026

Deliverables

- Submit Application function functionality
- Application status viewing
- Shortlist/Reject Application
- NQF Qualification Integration (SA-QA Aligned)
- Skill Tag Standardization (SA – Aligned dataset)
- In-app notifications for application lifecycle events
- Full application submission with CV integration

Spring 4

Duration: 1 weeks (13 May – 18 April 2026)

Milestone	Deadline
Sprint 4 Planning and Task Allocation	13 May 2026
Progress Update Meeting	15 May 2026
Progress Update Meeting	16 May 2026

Milestone	Deadline
Code Progress Review and Sprint Progress Check	17 May 2026
All Sprint 3 Tasks Completed	18 May 2026

Deliverables

- Analytics dashboard for admins and providers
- Graphical/tabular application volume reports
- Sector based placement success rate analysis
- Custom analytics View
- Customizable dashboard views
- Automated opportunity matching algorithm using NQF and skills dataset
- CVS/PDF export functionality
- Email notification system

Overall Project Timeline Summary

Sprint	Focus Area	Completion Deadline
Sprint 1	Authentication, Roles, Opportunity posting and Approval	13 April 2026
Sprint 2	Profiles, Applications, Testing, UML Diagrams	19 April 2026
Sprint 3	Application Workflow, Notifications, NQF & Skills Standardization	10 May 2026
Sprint 4	Analytics, Reports and Auto matching, Email Notification	18 May 2026

Key Deliverables by Category

Authentication and Security

- Google third party login/register
- JWT-Token authentication
- Role-based route protection middleware

User Management

- Master admin (first registered user)
- Role assignment (Admin, Provider, Applicant)
- Profile creation and editing (with CV upload)

Opportunity Management

- Create opportunities
- Admin approval/rejection workflow
- Provider views own opportunities
- Resubmit rejected opportunities

Application Management

- Submit application using profile and CV
- Provider shortlist/reject applications
- Track application statuses
- View all applications per opportunity

Data Standards

- NQF-aligned qualification (SA-QA)
- Standardized skills dataset (SA – aligned)
- Qualification and skills matching logic

Notifications

- In-app notifications for application received, shortlisted and rejected
- Email notifications for application received, shortlisted and rejected

Analytics and Reporting

- Application volume report
- Placement success rates by sector
- Custom Analytics report
- Customizable dashboard filters
- CVS/PDF export

Quality Assurance

- Unit test with Jest
- Minimum 65% code coverage
- GitHub Actions CI pipeline
- Issue tracking on GitHub

Team Roles and Responsibilities

Team member	Responsibilities
Makovha Thendo (2585063)	Role assignment, middleware, edit profile, UML diagrams, NQF and skills standards, CV upload, and custom analytics views report
Dikeledi Mokoatle (2769106)	Registration, Google auth, default user role assignment, create/edit profile, shortlist applications, application volume report
Ozuko Mabongo (2768437)	Login, JWT, route protection middleware, view opportunities (provider), reject applications, placement success reports
Tlhalefo Sebaeng	Dashboard, create opportunity form, store opportunities, deployment (Azure), testing, application status view, in-app and email notifications
Musawenkosi Nzuza	Approve/reject opportunities, resubmit opportunities, submit applications, application linking, auto matching

Risks and Mitigation Strategies

Risk	Mitigation
Late deployment to Azure	Team testing of deployment before deadlines

Risk	Mitigation
Incomplete test coverages	Jest code coverage tool runs on every pull request, minimum 65% is required.
Merge Conflicts	Feature branches and pull requests with peer reviews
Missing deadlines	Daily scrum meeting on MS Teams and clear task ownership
UI consistency issues	Figma designs agreed before implementation